

Report on LearnHub System Capabilities and Tools

1. Introduction

LearnHub is a comprehensive e-learning platform designed to facilitate remote education by providing an interactive environment that connects learners and instructors. The system was developed using modern software engineering practices, applying Clean Architecture with support for the CQRS pattern and the MediatR library to organize data flow, improve scalability, and enhance maintainability.

2. System Capabilities

LearnHub offers a wide range of features tailored to different user roles (students, instructors, administrators):

User Management: Registration, authentication, and role-based authorization (student, instructor, admin).

Instructor Requests: Users can apply to become instructors.

Course Management: Add, edit, delete courses with category classification and cover images.

Course Review Workflow: Instructors submit courses for review; admins approve or reject.

Course Access: Students can browse and enroll in approved courses.

Payment Integration: Secure course purchases via Stripe.

Enrollment Tracking: Students can follow lessons and track progress.

Course Ratings: Students can rate courses, update ratings, or remove them.

Instructor Profiles: Public instructor pages showcasing their courses.

Search Functionality: Search courses by title, description, category, or instructor name.

3. Programming Tools and Technologies Used

The system leverages modern frameworks and libraries:

ASP.NET Core MVC – backend framework for controllers and views.

Entity Framework Core – ORM for database operations.

SQL Server – relational database management system.

Identity Framework – user authentication and role management.

MediatR – implements CQRS and decouples command/query handling.

CQRS (Command Query Responsibility Segregation) – separates read and write operations for performance and clarity.

Clean Architecture – layered structure (Domain, Application, Infrastructure, Web) for maintainability.

Stripe API – payment gateway integration.

Bootstrap & Razor Views – responsive front-end design.

4. Comparison with Similar Systems

Feature	LearnHub	Udemy	Coursera
Local instructor support	Yes	Yes	Yes
Instructor upgrade requests	Yes	No	No
Clean Architecture applied	Yes	Not publicly stated	Not publicly stated
CQRS + MediatR support	Yes	No	No
Payment integration (Stripe)	Yes	Yes	Yes
Student course ratings	Yes	Yes	Yes
Category & course management	Yes	Yes	Yes
Open-source availability	Yes	No	No

Key Differentiator: LearnHub emphasizes modern architectural patterns (Clean Architecture, CQRS, MediatR), making it more flexible and easier to extend compared to commercial closed-source platforms.

5. Conclusion

LearnHub represents an advanced model of an e-learning platform developed according to best practices in software engineering. By applying Clean Architecture and the CQRS pattern with MediatR, the system achieves a robust, scalable, and maintainable structure. This makes it an excellent choice for a graduation project, demonstrating mastery of modern programming techniques and architectural design.